RECOVERY FINE-TUNING RECOVERS WHERE IT WAS TRAINED: DURATION- AND DOMAIN-DEPENDENT GAINS IN PRUNED TEXT-TO-AUDIO DIFFUSION

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ABSTRACT

Structured pruning of text-to-audio diffusion models is followed by recovery fine-tuning, whose benefit is usually certified by one score at one inference setting. Using the released pruned and fine-tuned AudioLDM-M checkpoints at two pruning severities (65% and 83% of U-Net parameters removed), we measure the paired recovery gain in text–audio alignment across clip duration and prompt domain, anchored by a shuffled-caption chance floor and the real audio of the same prompts. At 83% pruning, fine-tuning closes 63% of the pruned checkpoint's gap to real audio at its own 10.24 s fine-tuning duration but only 33% at 3.84 s, and nothing on held-out hip-hop captions; against the unpruned model, 82% versus 44% at 83% and 52% versus 8% at 65%. The duration dependence was pre-specified, replicates on a disjoint prompt set, survives a family-wise correction, is not a scorer artefact, grows monotonically over four durations, holds at the published sampler recipe, and is reproduced at both durations by a second scorer and by two event-level metrics outside the CLAP family. Recovery should be reported across operating points; lacking a matched dense fine-tuned control and a listening panel (one author's blinded listening agrees at the native point, not at 3.84 s), our claims concern evaluation, not mechanism.

Index Terms— text-to-audio generation, diffusion models, structured pruning, recovery fine-tuning, evaluation

1. INTRODUCTION

Structured pruning followed by recovery fine-tuning is the standard route to cheaper diffusion models [1–3], recently applied to text-to-audio (TTA) generation with AudioLDM [4, 5]. Recovery is judged by one aggregate number on the in-domain test set at one sampler setting, yet a generative model is used at an inference operating point: a clip duration, a prompt domain, a sampler configuration. Recovery fine-tuning is itself training at one operating point; nothing guarantees that its gain holds elsewhere.

We ask: does the gain of recovery fine-tuning over the pruned checkpoint depend on the operating point at which it is evaluated? We take the released pruned and fine-tuned AudioLDM-M-Full checkpoints of Singh et al. [5] at two pruning severities, train nothing, and measure the paired recovery gain in text–audio alignment across clip duration and prompt domain, read against a chance floor and a real-audio ceiling. Our findings:

- Duration. The recovery gain is several times larger at the fine-tuning duration (10.24 s) than at 3.84 s, at both severities, and grows monotonically over four durations at severity 2. The interaction was pre-specified, replicates on a disjoint prompt set, and is not a scale or scorer effect (the chance floor moves by at most 0.025 between durations).
- Domain. At matched duration the gain is present in-domain (AudioCaps) and absent or negative on held-out hip-hop/rap captions—at both severities at 3.84 s and at both durations at severity 2—where alignment after fine-tuning sits 0.02–0.03 above the chance floor, against 0.32 in-domain.
- Where the gain lives. Frame-level grounding shows the native gain spread uniformly over the clip; a crop analysis shows the short-duration deficit comes from generating short clips, not from scoring short excerpts.

2. BACKGROUND

Pruning and recovery in diffusion models. Structural pruning with fine-tuning recovery is established for image diffusion models [1–3] and self-supervised speech encoders [6, 7]. For TTA, Singh et al. [5] L1-prune the AudioLDM U-Net, fine-tune on AudioCaps for 10^6 steps and report in-domain FAD/KL at a fixed sampler; their fine-tuned pruned model even surpasses the unpruned one in-domain (FAD 1.57 vs. 3.95), a first sign of in-domain specialisation. All evaluate recovery at one operating point.

Evaluating text-to-audio generation. CLAP similarity [8] is the standard automatic proxy for text–audio alignment [9, 10]; FAD [11] is sensitive to sample size and embedding [12, 13]; event classifiers [14], human-aligned CLAP [15] and frame-level grounding [16] measure other axes.

Duration as an operating point. Latent audio diffusion conditions on the latent length and is trained at a fixed duration (10.24 s for AudioLDM [4, 17]), so duration is a natural operating point for a checkpoint fine-tuned at one; fine-tuning also trades in-distribution gains for out-of-distribution losses [18], motivating the domain axis.

3. EXPERIMENTAL DESIGN

3.1. Systems

The dense model is AudioLDM-M-Full [4] (U-Net stage multipliers $(1, 2, 3, 5)$, 415.96 M U-Net parameters), itself AudioCaps-fine-tuned by its authors. We study the two released pruning severities of [5]: severity 1, $(1, 2, 3, 1)$, 145.67 M parameters (−65.0%), and severity 2, $(1, 2, 1, 1)$, 71.08 M (−82.9%; 39% fewer multiply–accumulates [5]). At each severity we compare the pruned checkpoint (P) with the fine-tuned checkpoint (P+FT), the released result of 10^6 steps of AudioCaps recovery fine-tuning of P [5] (EMA weights). The released pruned checkpoints carry no usable EMA weights, so P is the released L1 channel selection applied to the dense EMA weights (bit-exact to the release at severity 1; at severity 2 the release differs in three decoder-seam tensors, so both conventions are evaluated: A′ primary, B′ sensitivity). "Recovery" names the fine-tuning stage, not an achieved restoration; we train no system. Two of the present authors released the evaluated checkpoints [5]; the operating points, batteries, estimands and gates below were specified independently.

3.2. Operating points and prompt batteries

Two durations: the native point (10.24 s, latent length 256; the fine-tuning duration) and a short point (3.84 s, latent length 96); a sweep adds 5.12 and 7.68 s (latent 128, 192) at severity 2. Two domains: in-domain AudioCaps test captions [19] and held-out hip-hop/rap captions from MusicCaps [20], a sub-domain with near-zero exposure in the recovery corpus ("music"; captions seven times longer, median 56.5 vs. 8 words, so the domain axis bundles content with caption style). Severity 2 uses 192 AudioCaps prompts (disjoint from all severity-1 prompts) and 64 music prompts (3 replicates at 3.84 s, one at 10.24 s); severity 1 uses 96 AudioCaps prompts for the pre-specified domain test, a matched 80-prompt subset for the duration comparison, and 64 music prompts at 3.84 s; the dense model is generated at both durations on both AudioCaps sets (matched controls, same noise). Generation uses DDIM [21], 50 steps, guidance 2.5, $\eta = 0$, single generation, EMA weights (off the published recipe, Sec. 5; a 64-prompt severity-2 check uses the published DDIM 200, guidance 3.5).

3.3. Paired protocol, scoring and anchors

Within each operating point the compared systems receive identical generation noise x_T per prompt (common random numbers); across durations the latent shapes differ, so duration contrasts are prompt-paired but not noise-paired. The primary scorer is fused CLAP (laion/clap-htsat-fused, rev. 365dea6e) [8], which repeat-pads 3.84 s audio to 10 s and centre-crops 10.24 s audio to 10 s; every (system, operating point) cell is scored with one fixed-order, fixed-seed call, part of the endpoint. Two anchors make absolute scores readable: a cell's chance floor, the mean cosine between each clip and the captions of the other prompts in its battery (a shuffled-caption baseline from the same embeddings), and the real-audio ceiling, the real AudioCaps clip of each prompt, band-limited to 16 kHz and scored under the identical convention at full length and as its first 3.84 s. The unit is the prompt (music replicates averaged first); intervals are 95% prompt-level percentile bootstrap intervals ($B = 10^4$). Finally, one author listened blind (opaque ids, sealed key, seeded draw) to 8 AudioCaps and 8 music prompts at severity 2: one listener, not a perceptual study.

3.4. Quantities and analysis plan

Let R be the mean per-prompt recovery gain, CLAP(P+FT) − CLAP(P), at a given (domain, duration); R_{nat}, R_{short}, R_{mus} denote AudioCaps-native, AudioCaps-short and music-short. The duration interaction is $J = R_{\text{nat}} - R_{\text{short}} = s(P{+}FT) - s(\text{P})$, where the duration response $s(\cdot)$ is a system's mean per-prompt score change from 3.84 to 10.24 s; J is paired per prompt, with pre-specified gate $\text{lo}_{95}(J) > 0$. The recovery ratio $\rho = R/(\text{ref} - \text{P})$ is the fraction of the pruned checkpoint's gap to a reference (dense, or the real audio of the same prompts) that fine-tuning closes. The domain contrast at matched duration is $R_{\text{short}} - R_{\text{mus}}$; the smallest effect size of interest is 0.025. "Pre-specified" means estimand, gate and prompt set were committed before any score was seen: the severity-1 domain test (the original hypothesis; failed), the severity-1 duration follow-up, the severity-2 replication with seam sensitivity (primary), the severity-2 music cell at 10.24 s and the secondary metrics. The FineLAP diagnostic, the severity-2 dense control, the duration sweep and the published-recipe check were registered after the primary result: committed, with their decision rules, before their own scores were seen, after the primary endpoint had been read. The severity-1 dense control, crop, rank-scale, Holm and anchor analyses and the author listening are post-hoc.

4. RESULTS

4.1. The recovery gain grows with clip duration

At the more severe operating point, recovery fine-tuning buys several times more alignment at its own duration than at 3.84 s. On the disjoint 192-prompt set, with estimands frozen before scoring, the interaction is clearly resolved, $J = +0.159$ (Fig. 1, Table 1). Fine-tuning raises CLAP alignment from 0.055 to 0.299 at the native duration ($R_{\text{nat}} = +0.244$) but only from 0.015 to 0.100 at 3.84 s ($R_{\text{short}} = +0.085$; resolved, not absent); the pruned checkpoint gains just $s(\text{P}) = +0.040$ from the longer clip, the fine-tuned one $s(P{+}FT) = +0.200$. The seam convention does not matter (B′: $J = +0.161$).

The effect is not an artefact of scale. The chance floor is close to zero and nearly duration-independent (Table 2: −0.048 to +0.020 across the AudioCaps cells, moving by at most 0.025 between durations for any system), so the interaction survives chance correction: $J_c = +0.191$ [+0.162, +0.220]. Real audio of the same prompts scores 0.274 at 3.84 s and 0.440 at 10.24 s ($s(\text{real}) = +0.167$ [+0.150, +0.184]); against that ceiling fine-tuning closes $\rho = 63\%$ of the pruned checkpoint's gap at the native duration but 33% at 3.84 s.

The matched dense control separates the systems' duration response from the scorer's. At severity 1, on the same 80 prompts and under the same convention, the dense model's response $s(\text{dense}) = +0.150$ ($0.202 \rightarrow 0.352$) equals the pruned checkpoint's, $s(\text{P}) = +0.149$ (paired difference −0.001 [−0.056, +0.055]), while the fine-tuned checkpoint responds more, $s(P{+}FT) = +0.193$ (excess over dense

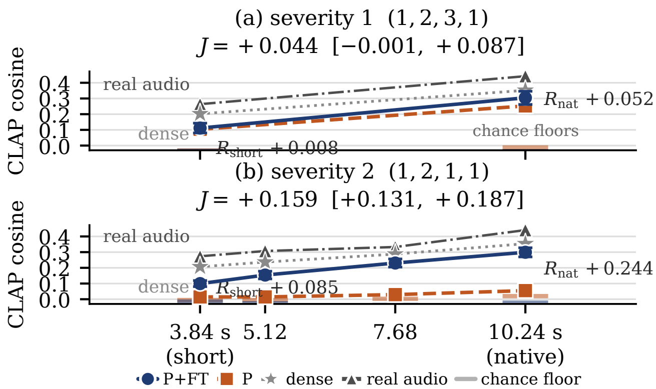

Fig. 1. The recovery gain grows with clip duration. Mean CLAP cosine of the pruned (P, dashed) and fine-tuned (P+FT, solid) checkpoints at the short and native points, (a) severity 1, and (b) severity 2 with the two sweep durations; whiskers: 95% CI of the paired gain R about the P+FT mean. Anchors: real audio of the same prompts (triangles), each cell's chance floor (ticks) and the matched dense control (stars; same prompts and scoring convention).

$+0.043$ [-0.020, $+0.109$]). The recovery gain itself is small there (Table 1), and part of the pruned checkpoint's raw response is a rise of its own floor ($+0.019$; the fine-tuned checkpoint's floor does not move), so on chance-corrected scores the severity-1 interaction is resolved, $J_c = +0.066$ [$+0.020$, $+0.111$]. The severity-2 control reads the same on its own prompts ($0.207 \rightarrow 0.354$): $s(\text{dense}) = +0.147$ [$+0.117$, $+0.177$], against which the pruned checkpoint responds -0.107 [-0.137, -0.076] less and the fine-tuned checkpoint $+0.053$ [$+0.017$, $+0.089$] more (paired).

The gain grows monotonically with duration. A sweep of P, P+FT and dense over 5.12 and 7.68 s on the same prompts (Fig. 1b) gives $R = +0.085$, $+0.139$, $+0.201$, $+0.244$ at 3.84, 5.12, 7.68, 10.24 s, every step resolved ($+0.054$ [$+0.032, +0.077$], $+0.062$ [$+0.037, +0.087$], $+0.043$ [$+0.014, +0.073$]): by the pre-specified shape rule the gain is monotone increasing. The pruned checkpoint is flat to 5.12 s (-0.000) while P+FT rises at every step; ρ_{dense} climbs 44%, 63%, 78%, 82% (Table 2); ρ_{real} dips at 10.24 s only because the real clip itself gains $+0.108$ once it fills the scorer's 10 s window without repeat-padding. To the blinded listener the gain is audible at 10.24 s (P+FT preferred 6/8, P mostly heard as noise) but not at 3.84 s (0/8; both heard as noise), so the short gain CLAP resolves is below what one ear picks up.

Severity 1 is underpowered, not negative: its raw-scale interval narrowly includes zero (the follow-up was powered for $J \geq 0.065$ at $n = 80$), but it is directionally consistent on every scale (post-hoc): the median per-prompt interaction is $+0.051$ [$+0.012$, $+0.077$] and the second scorer gives $J_{\text{HC}} = +0.075$ [$+0.012$, $+0.137$]. A Holm correction over all 19 contrasts, sweep steps and published-recipe check included, leaves every severity-2 conclusion unchanged ($p < 10^{-4}$; last sweep step $p = 0.003$); at severity 1, R_{nat} ($p = 0.016$) and J ($p = 0.052$) do not survive, so the severity-1 duration effect rests on the severity-2 replication.

Table 1. Recovery gain in CLAP alignment: mean cosine per checkpoint, paired gain R with 95% CI, and fraction W of prompts on which P+FT beats P. Duration rows give s(P), $s(P{+}FT)$ and, in the R column, J (domain contrasts two-sample; n as in Sec. 3.2). †: CI excludes 0; ‡: gate passes.

Setting	P	P+FT	R [95% CI]	W
Severity 1, (1, 2, 3, 1)				
AudioCaps 3.84 s	0.104	0.111	+0.008 [-0.023, +0.039]	0.44
AudioCaps 10.24 s	0.253	0.304	+0.052† [+0.009, +0.093]	0.64
music 3.84 s	0.117	0.023	−0.094† [-0.124, -0.065]	0.20
duration s; J	+0.149	+0.193	+0.044 [-0.001, +0.087]	
domain 3.84 s (n=96)			+0.092† [+0.054, +0.131]	
Severity 2, (1, 2, 1, 1)				
AudioCaps 3.84 s	0.015	0.100	+0.085† [+0.066, +0.105]	0.72
AudioCaps 5.12 s	0.015	0.154	+0.139† [+0.115, +0.164]	0.79
AudioCaps 7.68 s	0.029	0.231	+0.201† [+0.175, +0.227]	0.89
AudioCaps 10.24 s	0.055	0.299	+0.244† [+0.215, +0.273]	0.87
music 3.84 s	0.005	0.014	+0.009 [-0.013, +0.032]	0.48
music 10.24 s	0.089	0.094	+0.005 [-0.028, +0.039]	0.53
duration s; J	+0.040	+0.200	+0.159‡ [+0.131, +0.187]	
domain 3.84 s			+0.076† [+0.047, +0.105]	
domain 10.24 s			+0.239† [+0.195, +0.283]	

4.2. At matched duration the gain is domain-specific

The gain does not transfer off the fine-tuning domain, at either duration. At severity 2 and 3.84 s it is resolved in-domain ($R_{\text{short}} = +0.085$) and null on music ($R_{\text{mus}} = +0.009$; $W = 0.48$), a matched-duration domain contrast of $+0.076$; at the native duration the music contrast is likewise null ($+0.005$; $W = 0.53$), so the large native gain is confined to the fine-tuning domain (domain contrast $+0.239$; music duration interaction -0.004). At severity 1 the pre-specified domain test (96 prompts) found no in-domain gain at 3.84 s ($R_{\text{AC}} = -0.002$ [-0.027, $+0.021$]) and a negative music contrast ($R_{\text{mus}} = -0.094$; domain contrast $+0.092$).

The chance floor reads the two severities as one statement (Table 2): after fine-tuning, alignment on the hip-hop captions is 0.022, 0.018 and 0.033 above chance (severity 1 at 3.84 s; severity 2 at 3.84 and 10.24 s), against 0.115 and 0.321 in-domain at severity 2. Whether that registers as a "penalty" depends on whether the pruned checkpoint had music alignment to lose: the 65%-pruned one did (0.061 above chance; chance-corrected $R_{\text{mus}} = -0.040$ [-0.060, -0.020]), the 83%-pruned one did not (0.018). The scorer's null is not a perceptual tie: the listener preferred P+FT on 8/8 music pairs and heard it as music on 5/8 (P 1/8), though neither followed the long captions (1.4 and 2.5 of 5); musicality without alignment is invisible to a text–audio scorer. The null is near chance for both checkpoints (no real hip-hop reference was available for a ceiling) and not a caption-length effect: within AudioCaps the native gain is uncorrelated with caption length (Spearman $\rho = +0.04$ [-0.12, $+0.18$]), and no caption is truncated by the conditioner—though 47% of the music captions exceed CLAP's 77-token pre-training length, a caption-style covariate inseparable from content.

4.3. Where in the clip the gain lives

Two analyses locate the native-duration gain. First, a prospectively frozen diagnostic scores the native clips with FineLAP [16], a frame-level language–audio grounding model (EAT encoder; not CLAP-derived) reporting per 0.16 s frame how strongly the requested event is grounded (110/49 eligible

Table 2. Anchors and recovery ratios (AC = AudioCaps; n as in Table 1): chance floor of the P/P+FT cells, real audio of the same prompts, and the fraction ρ of the pruned checkpoint's gap to dense and to real audio closed by fine-tuning (95% CI, %). Statuses as in Sec. 3.4.

Setting	floor P/P+FT	real	ρ_{dense} [%]	ρ_{real} [%]
sev. 1 AC 3.84 s	$-0.031/-0.033$	0.264	8% [−30, 36]	5% [−16, 24]
sev. 1 AC 10.24 s	$-0.012/-0.036$	0.442	52% [11, 103]	27% [6, 46]
sev. 2 AC 3.84 s	$-0.005/-0.015$	0.274	44% [36, 53]	33% [26, 39]
sev. 2 AC 5.12 s	$-0.005/-0.025$	0.307	63% [52, 74]	48% [40, 55]
sev. 2 AC 7.68 s	$+0.003/-0.037$	0.333	78% [68, 88]	66% [59, 74]
sev. 2 AC 10.24 s	$+0.020/-0.022$	0.440	82% [72, 92]	63% [56, 71]
sev. 1 music 3.84 s	$+0.055/+0.001$	–	–	–
sev. 2 music 3.84 s	$-0.013/-0.004$	–	–	–
sev. 2 music 10.24 s	$+0.070/+0.061$	–	–	–

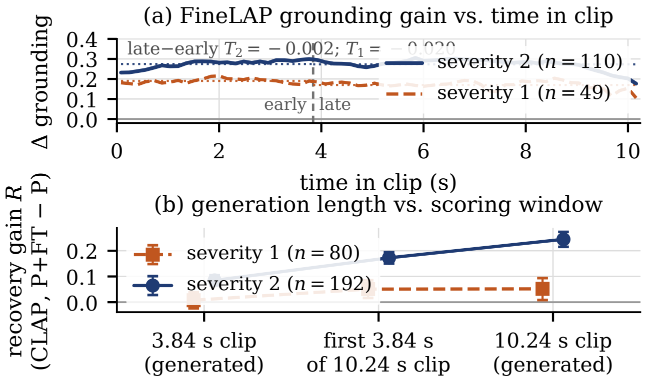

Fig. 2. (a) FineLAP frame-level grounding gain (P+FT − P) vs. time in the 10.24 s clip : uniform, not back-loaded. (b) R (95% CI) on the generated 3.84 s clip, the first 3.84 s of the 10.24 s clip and the full clip: a generation-length, not scoring-window, effect.

prompts at severities 2/1, outcome-blind). Fine-tuning raises grounding across the whole clip (severity 2: semantic mass +0.27 [+0.22, +0.33]; both severities; seam-robust), and the gain in the first 3.84 s equals the gain in the remainder: $T = -0.002$ [-0.024, $+0.020$] (Fig. 2a). Second, a post-hoc crop analysis scores the first 3.84 s of each 10.24 s generation under the same convention against the separately generated 3.84 s clips (Fig. 2b). The crop carries a large part of the native gain: at severity 1 it equals R_{nat} and exceeds R_{short} by +0.043 [+0.002, +0.085]; at severity 2, $R_{\text{crop}} = +0.172$ [+0.150, +0.194], $R_{\text{crop}} - R_{\text{short}} = +0.087$ [+0.065, +0.110], $R_{\text{nat}} - R_{\text{crop}} = +0.072$: the deficit is a generation effect, not a scoring one.

4.4. Corroboration and pre-specified negatives

At the severity-2 native point Human-CLAP [15] (a CLAP fine-tuned on human ratings; automatic) gives $R_{\text{nat}} = +0.375$ [+0.340, +0.408] and $J = +0.185$; two pre-specified event-level metrics favour P+FT: KL to real references, 2.23 vs. 4.45 ($\Delta = +2.22$ [+1.93, +2.53]); PANNs top-10 captured labels, 1.46 vs. 0.60 ($\Delta = +0.86$ [+0.70, +1.02]). Rescored at 3.84 s against the same references cropped alike (post-hoc), both gains shrink—KL +0.66 [+0.42, +0.92], PANNs +0.19 [+0.06, +0.32]—so the interaction holds outside the CLAP family: $J_{\text{KL}} = +1.56$ [+1.19, +1.92], $J_{\text{PANN}} = +0.67$ [+0.49, +0.86]. At the published sampler recipe (DDIM 200, guidance 3.5; single generation) on the first 64 prompts the interaction holds, $J = +0.184$ [+0.126, +0.243] (pre-specified gate; the frozen recipe on the same prompts gives +0.168, difference +0.016 [-0.040, +0.072]). Off the primary scorer (post-hoc), Human-CLAP, KL and PANNs capture all show R rising at every step of the sweep, the first two steps resolved and the last (7.68 → 10.24 s) not: Human-CLAP is flat there (+0.004 [-0.033, +0.041]), KL +0.27 [-0.04, +0.57], PANNs +0.10 [-0.07, +0.27].

Three pre-specified hypotheses failed and are kept: (i) that fine-tuning trades in-domain for out-of-domain alignment: it requires an in-domain gain at 3.84 s, which severity 1 lacks ($R_{\text{AC}} \approx 0$), so the music loss is real but the trade is not; (ii) the severity-1 music penalty did not replicate (Human-CLAP alone shows -0.037 [-0.068, -0.005] at severity 2); (iii) a "late allocation" account of the duration effect is rejected by FineLAP ($T \approx 0$). Finally, P+FT is not restored to dense: at 10.24 s the dense model leads it by +0.048 [-0.000, +0.096] at severity 1 and +0.055 [+0.021, +0.088] at severity 2 (+0.107 at 3.84 s).

5. DISCUSSION AND LIMITATIONS

Across severities, scorers and metrics, the recovery gain tracks the operating point of the fine-tuning itself: growing monotonically toward the fine-tuning duration, confined to the fine-tuning domain, heterogeneous across prompts. At 65% pruning P still responds to duration like dense and fine-tuning adds a modest in-domain increment. At 83% pruning it is barely above chance at either duration (0.020 and 0.035 above the floor; to a listener, mostly noise), and fine-tuning restores a duration response above the dense model's—only in the fine-tuning domain and increasingly toward the fine-tuning duration, consistent with recovery acting in part as further in-domain, native-duration specialisation. Recovery studies should report both operating points, floors, paired gains and their interaction.

Limitations. One case study (one model family, two severities, two domains). Mechanistic attribution is blocked: the matched control (a dense model fine-tuned identically) no longer exists and the public dense text-fine-tuned release is not equivalent, so the dependence cannot be attributed to pruning rather than to fine-tuning in general. Primary inference rests on CLAP-family scorers; the only listening is one author's, blinded but informal ($N = 1$), and it disagrees with CLAP at 3.84 s. Sampler settings differ from the published recipe; the interaction is reproduced at its DDIM 200 / 3.5 on a 64-prompt subset, best-of-3 selection was not. The sweep stops at the fine-tuning duration, so it cannot separate "largest at the training duration" from "larger for longer clips". One held-out sub-genre; the severity-1 duration test was underpowered.

6. CONCLUSION

Recovery fine-tuning of pruned AudioLDM-M recovers much at its own operating point and much less away from it: at 83% pruning it closes 63% of the gap to real audio at the fine-tuning duration, 33% at 3.84 s and none on held-out music. Audio: gbibbo.github.io/audioldm-modality-swap-pruning.

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